

WEEK 9 PROGRESS REPORT

Full AI-Guided Fuzzing Experiments

July 30, 2026

1. Week 9 Objectives

This week I completed the full AI-guided fuzzing experiment phase. The goals were to run the experiments, collect and validate the result sets, and begin a fair preliminary comparison across multiple language models.

2. Experimental Setup

- 24 target programs covering validators, parsers, configuration formats, arithmetic, and small protocol grammars.
- 3 independent trials per target and 500 AI-generated inputs per trial.
- 36,000 target executions per model and 144,000 AI-guided executions across the four official models.
- All models used the same target-aware prompt, response parser, safety validator, execution engine, and coverage measurement workflow.

Model	MindRouter model ID	Inputs	Status
Qwen 3.5 122B	qwen/qwen3.5-122b	36,000	Completed
GPT-OSS 120B	openai/gpt-oss-120b	36,000	Completed
Qwen 3.6 35B	qwen/qwen3.6-35b	36,000	Completed
Gemma 4 31B	google/gemma-4-31b	36,000	Completed

3. Result Validation

I validated each model archive before analysis. Every official model contains 72 JSONL result files and 72 metadata files, representing 24 targets multiplied by 3 trials. Every result file contains exactly 500 valid JSON records, giving 36,000 inputs per model. The metadata reports all groups as completed, and ZIP integrity checks passed.

- Verified official result total: 144,000 inputs.
- Verified per-model structure: 24 targets, 3 trials, 500 inputs per group.
- No invalid JSONL records or corrupted ZIP entries were found.

4. Operational Issues and Recovery

Qwen 3 32B was initially selected, but persistent MindRouter HTTP 502 failures left 3 of 72 groups incomplete. I preserved that attempt as an excluded side experiment and replaced it with Qwen 3.6 35B. Qwen 3.6 later recovered one failed group on retry. Gemma recovered four incomplete groups on retry after malformed JSON and temporary backend, DNS, and connection failures. No partial model was mixed into the official equal-budget comparison.

5. Preliminary Model Comparison

Model	Avg. coverage	Valid inputs	Error rate	Diversity	Generation
Qwen 3.6 35B	74.43%	16,519	54.11%	84.28%	1.52 h
Qwen 3.5 122B	74.14%	15,364	57.32%	88.33%	3.72 h
GPT-OSS 120B	73.40%	12,483	65.33%	86.84%	6.53 h
Gemma 4 31B	73.36%	15,763	56.21%	81.95%	2.91 h

Coverage is the mean union of covered source lines within each 500-input target/trial group. Diversity is the percentage of unique inputs across the three trials of each target.

6. Main Findings

- Qwen 3.6 35B provided the strongest efficiency-quality balance. It achieved the highest average coverage and valid-input count while completing generation in 1.52 hours.
- Qwen 3.5 122B produced the highest cross-trial diversity at 88.33%, but required 3.72 hours of generation.
- GPT-OSS 120B produced the highest error-triggering rate at 65.33% and the widest exception-type set, but it was the slowest model at 6.53 hours.
- Gemma 4 31B remained competitive in coverage and completed the final dataset after retrying temporary generation and network failures. The coverage spread across the four models was only 1.07 percentage points. Therefore, the preliminary result is not that one model dominated coverage, but that Qwen 3.6 reached comparable or slightly better coverage at substantially lower generation time.

7. Preliminary Context Against Week 8 Random Fuzzing

The Week 8 random baseline also used 36,000 executions. It produced 4,113 valid inputs (11.43%) and 31,887 error inputs (88.58%). In Week 9, the AI models produced between 12,483 and 16,519 valid inputs (34.68% to 45.89%). This supports the expectation that AI guidance can generate more semantically valid test cases. However, a higher target exception rate is not automatically evidence of better bug discovery, because many exceptions are expected validation failures.

8. Deliverables and Status

Week 9 deliverable	Status
Complete AI-guided results	Completed and validated
Preliminary comparison	Completed

9. Next Step

The next phase is to produce the final normalized random-versus-AI comparison by target and trial, add statistical summaries and charts, analyze unique failure behavior, and convert the preliminary findings into the evaluation section of the research paper.